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Robust securable economic operation of grid-connected renewable integrated energy system with hydrogen storage and electric and fuel cell vehicles

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The economic operation of the smart distribution network based on voltage security and environmental indicators considering renewable integrated energy systems based on hydrogen storage and electric vehicles is presented in this study. Renewable solar, bio-waste, and wind resources are used in integrated system. Battery-based electric vehicles, plug-in hybrid electric cars, and fuel cell-based electric vehicles are present as mobile storage in the integrated system. The weighted sum of the voltage profile, operating cost, and environmental pollution are minimized for a deterministic model. This problem includes power flow constraints, security and operation limitations, and the operating formulation of renewable units, hydrogen storage, various electric vehicles, and the power and hydrogen balancing in the integrated system. Fuzzy decision-making technique obtains a compromise point. In the following, a linear formulation for the aforementioned study is obtained. The proposed design has uncertainty parameters such as price of energy, renewable power rate, load, charge/discharge rate, and initial/final energy of electric cars. To achieve a robust solution to the prediction error of uncertainties, this paper uses adaptive robust optimization to appropriately formulation the aforementioned uncertain quantities. Finally, the findings indicate the suggested design's potential for improving the environmental, technical and economic conditions of the smart distribution grid with energy management of the aforementioned integrated unit. Thus, the power management of stationary and mobile storage devices in the renewable integrated system is able to improve the security of voltage, operational, economic and environmental status of the smart distribution system by about 42.5%, 43.9%-57.6%, 26% and 62%, respectively, in the worst case scenario compared to network load flow analysis.

Keywords Eco-securable operation, Electric vehicles, Hydrogen storage, Integrated energy system, Renewable sources, Smart distribution network

List of symbols

Variables

$\hat{E}_A, \hat{E}_D$	Uncertainty variable of initial and final energy (MWh) for the integration of battery-based electric vehicles (BEVs) and plug-in electric vehicles (PHEVs)
EC	Purchasing energy cost from the upstream grid ($)
E_{EV}, E_{HT}, E_{ET}	Stored energy in batteries of BEVs and PHEVs, hydrogen tanks, and fuel cell-based electric vehicles (FCEVs) in MWh
$\hat{E}_I, \hat{E}_F$	Uncertainty variable for initial and final energy (MWh) for FCEVs
EM	Environmental pollution (ton)
$H_{DIS}^{fcev}, H_{CH}^{fcev}$	Hydrogen power in FCEVs for discharging and charging performance (MW)
$H_{Dis}^{hs}, H_{CH}^{h.s}$	Hydrogen power in hydrogen tank for discharging and charging performance (MW)
H_{FC}, H_{EL}	Hydrogen power of fuel cell and electrolyzer (MW)
OF	Objective function

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$\hat{P}_C, \hat{Q}_C$	Uncertainty variable for active (MW) and reactive (MVAr) load
P_{CH}, P_{DIS}	BEVs and PHEVs aggregation power in for charging and discharging mode (MW)
P_{DL}, Q_{DL}	Active (MW) and reactive (MVAr) power of the line in the distribution network
P_{DS}, Q_{DS}	Active (MW) and reactive (MVAr) power of substations in the distribution network
P_{FC}, P_{EL}	Active power in fuel cell and electrolyzer (MW)
P_I, P_R	Active power in integrated energy system (IES) and renewable source (MW)
$V, \Delta V$	Voltage magnitude and deviation of voltage (p.u.)
VD	Voltage profile function (p.u.)
χ_W	Worst security index (p.u.)
$\hat{\varphi}_R$	Uncertainty variable for renewable power rate
σ	Voltage angle (radians)
$\hat{\rho}$	Uncertainty variable for energy price ($/MWh)
$\hat{\kappa}_{CR}, \hat{\kappa}_{DR}$	Uncertainty variable for the charging and discharging rates of BEVs and PHEVs aggregation in (MW)
$\hat{\upsilon}_{CR}^{fcev}, \hat{\upsilon}_{DR}^{fcev}$	Uncertainty variable for charging and discharging rates of FCEVs aggregate in (MW)
$\Delta V^+, \Delta V$	Positive and negative component of voltage deviation (p.u.)

Parameters

A	Bus and line intersection matrix in distribution network
E_A, E_D	Initial and consumed energy required in the integration of BEVs and PHEVs (MWh)
EC^{min}, EC^{max}	Minimum and maximum distribution network operating costs ($)
E_I, E_E	Initial and consumption energy required by FCEVs (MWh)
EM^{min}, EM^{max}	Minimum and maximum amount of environmental pollution (ton)
$E_{HT}^{in}, E_{HT}^{lo}, E_{HT}^{up}$	Initial energy, lower and upper amount of energy in tank of hydrogen (MWh)
P_C, Q_C	Load power in active (MW) and reactive (MVAr) term
P_{EL}^{up}, P_{FC}^{up}	Electrolyzer and fuel cell capacity (MW)
S_{DL}^{up}, S_{DS}^{up}	Size (capacity) for line and post in distribution network (MVA)
R, X	Line resistance and reactance in distribution network (p.u.)
VD^{min}, VD^{max}	Lower and upper amount of voltage profile function (p.u.)
V^{up}, V^{lo}	Upper and lower voltage magnitude (p.u.)
α_{DL}, β_{DL}	*Line conductance and suspensivity in distribution grid (p.u.)*
φ_R	Renewable power rate
ρ	Energy price ($/MWh)
κ_{DR}, κ_{CR}	Discharging and charging rates of BEVs and PHEVs (MW)
η_{DIS}, η_{CH}	Discharging and charging efficiency of BEVs and PHEVs
η_{EL}, η_{FC}	Electrolyzer and fuel cell efficiency
$\varpi_{EC}, \varpi_{VD}, \varpi_{EM}$	Weighting factors
$\upsilon_{CR}^{fcev}, \upsilon_{CR}^{fcev}$	FCEV charging and discharging rates (MW)
$\upsilon_{CR}^{hs}, \upsilon_{CR}^{hs}$	Hydrogen tank charging and charging rate (MW)
$\zeta_{CO2}, \zeta_{NOx}, \zeta_{SO2}$	CO2, NOx and SO2 emission coefficients

Indices

b, h	Bus, and operating hours
j	Auxiliary index corresponding to bus
k	Side in a regular polygon
p, p-1	Poor bus, upstream bus for poor bus

In this paper, the optimal operation constrained by different indicators in a smart distribution network in the presence of integrated energy systems is presented. The details of the design are given below.

Motivation

Integrated energy systems (IESs) are adopted for integrating and coordinating energy sources and storage devices, as well as producing and storing energy in distribution grid[1]. Aiming to diminish pollutant emissions, renewable energy sources (RESs) such as wind system (WS), photovoltaics (PV), and bio-waste systems (BS) in IESs play a significant use. Due to the uncertainty of RES output power, there are power fluctuations in RESs[2]. Therefore, to compensate for this issue, energy storage devices are used in IESs[3]. Of course, the use of storage devices is not only to deal with power intermittency of RES, but also with optimal energy management of storage devices, various technical and economic goals can be achieved in IESs and the network[3]. Electric vehicles (EVs) have also been emerged as a means of diminishing pollutant amount released to the environment. EV types are: hybrid EV (HEV), battery-based EV (BEV), plug-in HEV (PHEV), and fuel cell-based EV (FCEV)[4,5]. In the first type, the vehicle battery is charged through the kinetic energy available in the EV's motion. However, in the second and third types, EVs are connected to the power system to supply the battery and receive significant energy from the grid. In the fourth type, the EV fuel is hydrogen[5]. Today, in order to improve environmental conditions, hydrogen production is provided by electrolyzer connected to renewable sources or the power system. Although EVs appear as a load in the grid, with their optimal energy management, EVs can be activated as mobile storage devices[4]. Therefore, it is expected that this boosts the economic and technical objectives of the grid and IES. By implementing an energy management system (EMS) in a distribution grid with IESs, steps can be taken to improve the environmental, economic, and technical conditions of the grid. In EMS, power supplies and storage are generally managed by the IES operator. So that this operator achieves the optimal unit commitment for the mentioned elements by receiving the capacity data of the supplies and storage equipment[6]. Note however that the aforementioned operator has bidirectional communication with the distribution system operator (DSO). So,

the energy scheduling of the IES is accompanied by the estimation of the technical limits of the network and the demands of the DSO. Therefore, it is predicted that with optimal energy scheduling in distribution system with renewable IESs based on mobile and stationary storage devices, optimal economic, environmental, security and operational goals can be achieved.

Research background

Numerous studies and efforts have focused on optimizing the operation and energy management of IESs within distribution networks. For instance, Ref.[7] introduces a multi-time-scale game-theoretic optimization framework for Park-level IESs, incorporating various demand response models. In the day-ahead phase, an optimization scheduling model is developed, leveraging a detailed incentive-based demand response mechanism. The uncertainty associated with predicted distributed renewable energy outputs and diverse energy loads is captured using fuzzy chance-constrained programming. Similarly, Work in[8] proposes a synergistic planning approach for IESs equipped with hydrogen storage (HS). This model integrates the use of both electrical and thermal energy to significantly reduce carbon emissions and enhance energy efficiency. An electric-thermal coupling model for the hydrogen energy storage unit is formulated, alongside an optimization method that considers the dual utilization of electrical and thermal resources. In[9], a low-carbon economic operational model is presented, which accounts for the variable load characteristics on the demand side. It incorporates critical systems such as combined heat and power (CHP), a carbon capture system (CCS), and power-to-gas (P2G). This study explores the integration of source-side and demand-side scheduling to improve system operation comprehensively. Work in[10] develops a data-driven two-stage distributionally robust optimization (DRO) model to manage uncertainties in RESs alongside integrated demand response. This model employs a blended uncertainty set based on the 1-norm and ∞-norm to sidestep complex probability density functions. A nonlinear cooperative optimization model for regional IESs is introduced in[11], emphasizing multi-region energy sharing and multi-energy storage. A two-tier optimization strategy is proposed, with the upper tier focusing on renewable energy and storage capacity allocation, while the lower tier concerns operational dispatch. To tackle challenges related to robustness, economy, environmental protection, and efficiency, Ref.[12] proposes an IES group (IESG) framework through energy sharing and carbon transfer. This framework uses a two-stage distributionally robust optimization (TSDRO) model developed with kernel density estimation (KDE) and the Wasserstein metric. The study includes modeling projected power errors for wind turbines and photovoltaics through KDE-based probability density functions. Ref.[13] investigates the factors contributing to the integrated benefit (IB) index of IESs along with optimal energy supply strategies by developing a two-stage planning and operational optimization model.

Focusing on smart distribution networks (SDNs), Ref.[14] examines energy management strategies that integrate hydrogen storage and renewable energy sources. The study's objective function minimizes operational costs while improving flexibility, reliability, and reducing energy losses. The analysis is subject to AC optimal power flow equations, network reliability constraints, and integrated system models. Ref.[15] establishes a conventional multi-regional IES (MIES) model constrained by global carbon emission limits. This approach is framed as a partially observable Markov decision-making process (POMDP) and implements an enhanced multi-agent deep deterministic policy gradient (MADDPG) algorithm. The centralized training but decentralized execution (CTDE) method substantially improves multi-agent stationarity. In[16], a Stackelberg game-based simulation model for energy utilization is introduced. It incorporates integrated demand response (IDR) with user-side flexible resources (FRs) while considering consumer comfort costs. Meanwhile, Ref.[17] focuses on developing a competitive yet collaborative optimization strategy that accounts for interactions between supply and demand. It integrates carbon trading incentives on the supply side alongside uncertain user behavior in demand response. The framework further accommodates stakeholders' roles and energy-carbon market mechanisms, incorporating psychological factors affecting consumers' responses. Lastly, Ref.[18] outlines a synergy planning approach for Regional IESs (RIES), emphasizing cascade energy utilization while addressing uncertainty. It introduces a dual-layer synergy planning model, where the higher layer targets station-level optimization while addressing interconnected operational challenges.

Ref.[19] proposes an IES operation strategy that combines the adiabatic compressed air energy storage (A-CAES) device and the integrated demand response (IDR) theory with the two-layer optimization model, and comprehensively considers the interaction between the planning and operation of two different time scales. The improved two-level optimal scheduling method in this paper takes IES operators and users as different subjects, and coordinates the output of multiple energy sources on the basis of meeting IES constraints to minimize the cost. In[20], the system operator is accountable for reducing the weighted sum of the voltage security index, energy loss, and energy cost of the distribution network. This problem is associated with the optimal power flow formulation, which considers the environmental limits and security of voltage in the distribution network, the renewable resource operation model and flexibility in the form of a virtual power plant, and the system's flexibility constraints. Refs.[21] and[21] discusses the simultaneous management of active and reactive power of a flexible renewable energy-based virtual power plant placed in a smart distribution system, based on the economic, operational, and voltage security objectives of the distribution system operator. Work of[22] introduces a scenario-based scheduling approach tailored for the smart grid. The problem is formulated with a dual-objective function that addresses both emissions and costs, focusing on the traditional unit commitment while incorporating electric vehicles (EVs) and renewable energy technologies (RETs). Meanwhile, study in[23] explores the effects of high penetration of renewable energy sources (RES) and EVs on generation scheduling and total system costs. A two-stage model is proposed: the first stage involves an aggregator framework for scheduling EVs based on anticipated day-ahead electricity market prices, while the second stage focuses on a market model designed to achieve optimal generation scheduling that integrates RES and EV considerations.

The work done in the research background is summarized in Table 1.

Ref	DSO objective				Model of (.) in IES			Uncertainty model
	Security	Operation	Environmental	Economic	BS	HS	EV	
[7]	×	✓	×	✓	×	×	×	Stochastic optimization
[8]	×	✓	×	✓	×	✓	×	
[9]	×	✓	✓	✓	×	×	×	
[10]	×	✓	×	✓	×	×	×	Distributionally robust optimization
[11]	×	✓	×	✓	×	×	×	Stochastic optimization
[12]	×	✓	✓	✓	×	×	×	Distributionally robust optimization
[13]	×	✓	✓	✓	×	×	×	Stochastic optimization
[14]	×	✓	×	✓	✓	✓	×	
[15]	×	✓	✓	✓	×	×	×	
[16]	×	✓	×	✓	×	×	×	
[17]	×	✓	✓	✓	×	×	×	
[18]	×	✓	×	✓	×	×	×	
[19]	×	✓	✓	✓	×	×	×	
[20]	×	✓	×	✓	×	×	×	
[21]	✓	✓	✓	✓	✓	×	BEV, PHEV	
[21]	✓	✓	×	✓	✓	×	BEV, PHEV	
[22,23]	×	✓	×	×	×	×	BEV, PHEV	
Current approach	✓	✓	✓	✓	✓	✓	BEV, PHEV and FCEV	Adoptive robust optimization

Table 1. Classification of contemporary research studies.

Research gaps

Research limitations for managing energy of IESs in distribution networks according to Table 1 and the research background are:

- The study of the operational and economic indicators of distribution grid with IESs has been carried out in most research. In the economic field, minimizing the cost of network operation has generally been considered. In the issue of operation, the status of the voltage profile and energy losses has generally been examined. However, the distribution network has other important indicators. Among the most important technical indices in distribution grid, especially the low-voltage distribution network, is voltage security. The length of the feeders in the distribution network is generally long. Therefore, the loads at the end of the feeder have a low voltage level. Therefore, if the loading increases at these points, voltage collapse may occur. However, IESs are located in consumption areas that can take steps to improve voltage security by delivering power into the network. Yet, it has rarely been considered in studies, such as in[21]. Nowadays, providing clean energy using environmentally friendly technologies has received much attention. This issue has been included in many studies. But in few studies, such as[9,12,13,15,17,19], the numerical amount of environmental pollution in the energy management system has been calculated. But to know the status of an indicator, it should be measured, which can be extracted using the mathematical model of the aforementioned indicator. In addition, in the aforementioned studies, the reduction of carbon dioxide (CO2) emissions has generally been considered. However, the supply of electrical energy from centralized power plants includes CO2, sulfur dioxide (SO2) and nitrogen dioxide (NOx) emissions. The simultaneous reduction of these emissions has been considered in few studies.
- In most studies, wind and solar sources has been considered in the IES. The bio-waste system (BS) can also be adopted for such applications. The input to the BS is environmental waste and its output is gas. Then it converts gas into electrical energy. Since environmental waste landfilling requires a large area, the use of BS is very effective in reducing environmental pollution in addition to energy production. It is also predicted that the BS in the IES or network could also be effective in improving the technical and economic conditions of network. Nonetheless, the use of this renewable source has been considered in few studies such as[14,21]. In addition, battery storage is another choice to be used in IES. High efficiency and power density are some features of the batteries, but their life cycle is not considerable and their capital cost is high. Also, access to their high capacity is difficult. To compensate for this issue, compressed air and hydrogen storage can be used in IES. Their investment cost is low compared to batteries, these storages have a long useful life, and they also have a desirable efficiency. This issue has been considered in few studies. In addition to storing electrical energy, hydrogen storage (HS) can also act as a power supply for hydrogen loads. This can play a role in increasing the efficiency of the aforementioned storage. Yet, HS was not addressed thoroughly in the literature, even though some papers like [8 and 14] focused on it. These studies have also used HS only as an electrical energy storage device.
- EVs are new consumers in the power system, and with their energy management, EVs can be used as mobile storage devices. Therefore, they can be used to improve the technical, environmental, and economic indicators. In[21], BEVs and PHEVs have been used for economic management of distribution grid by considering the voltage security constraint. So, according to the results of these references, EVs have been able to diminish the

energy losses and operating cost of network, improve the voltage profile, and obtain a stable voltage margin for the network. Another type of EV is FCEV, which can also be used as a hydrogen storage device and can be adopted alongside HS for electrical energy storage purposes. Nevertheless, FCEVs in energy management of distribution networks have received less attention.

- The energy scheduling scheme of IESs in the distribution grid is accompanied by various uncertainty parameters. In[7–20], the uncertainty for price of energy, load, and renewable power are generally considered. In addition to these parameters, the uncertainty parameters of EVs are modeled in[21]. In most of the research, stochastic optimization models the uncertainty parameters. Since the aforementioned scheme has a high number of uncertainty parameters, stochastic optimization needs tremendous scenarios to extract a trustable solution, which can extend the problem size and, consequently, computing time will be long. The aforementioned scheme is the operation problem; whose execution step is small. Thus, in some applications like real-time operation, the execution step reaches 5 min. Therefore, the low computational time in power system operation problems is of particular importance. However, stochastic optimization does not satisfy the aforementioned conditions. To compensate for this, some research such as [10 and 12] uses robust optimization to model uncertainties. This technique has only one scenario called the worst-case scenario. Therefore, the computing time in this technique is less than stochastic optimization. It is also able to extract a solution that is robust to the maximum prediction error of uncertainties. However, in [10 and 12], robust optimization is only used for load parameters, energy prices, and renewable resources, and modeling the uncertainties of different types of EVs is not considered.

Research contributions

The energy scheduling of SDNs with the contribution of renewable IESs is presented in this study to estimate the security, operational, economic, and environmental goals of DSO as shown in Fig. 1. IES has renewable resources such as WS, PV and BS. The stationary storage of HS type is used in IES. IES also includes mobile storage of BEV, PHEV and FCEV types. In the deterministic formulation of study, minimizing the weighted sum of the SDN operating cost, the voltage deviation function, and the SDN environmental emissions (including CO2, SO2 and NOx) is the objective function. Constraints contain AC optimal power flow model subject to the voltage security constraint in SDN, the operating formulation of the renewable units, HS and EVs in the form of IES. By changing the weight coefficients in the objective function, different optimal points are obtained. In this paper, the fuzzy decision making approach extracts a compromise point. In the next step, a linear approximation formulation for scheme is obtained. One of the objectives of this paper is to use robust optimization to fill the last research gap. This technique requires a linear model of a problem. Here, the linear programming of the design is expressed. The uncertainty parameters in this scheme include load, energy price, renewable energy generation rate, discharge/charge rate, and EVs initial/final energy. Adaptive Robust Optimization (ARO) provides a suitable modeling of the aforementioned uncertainties. This technique has only one scenario, which includes the worst-case value of the uncertainty parameters proportional to the objective function. Therefore, the aforementioned technique obtains the most robust energy management for IES, such that it is robust against the maximum prediction error of uncertainties. Finally, by making a comparison between the presented solution and the literature works, the following innovations are extracted for the proposed design:

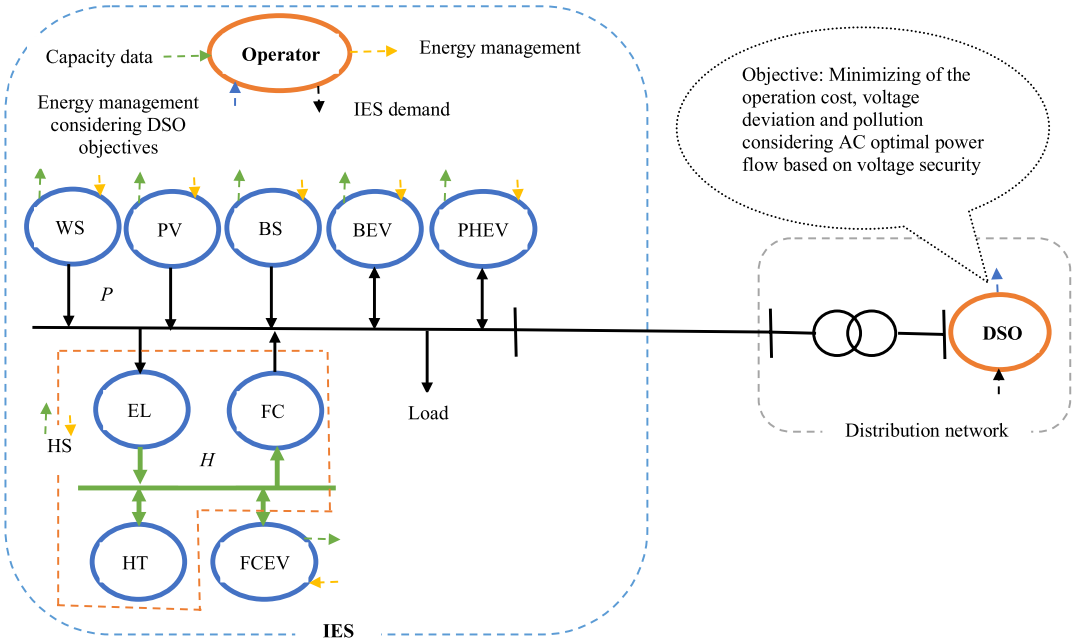


Fig. 1. Energy management of IESs according to DSO objectives.

- Implementation of an energy management system in a SDN in the presence of IESs in order to meet economic, environmental, operational and voltage security goals,
- Use of BEVs, PHEVs and FCEVs in IES as mobile storage to improve the economic, technical and environmental conditions of SDN,
- Modeling of renewable bio-waste source in IES to reduce environmental pollution while improving the economic and technical condition of the network,
- Modeling of HS in IES as an electrical energy storage and feeding hydrogen loads to increase the storage efficiency, and
- Simultaneous modeling of uncertainties related to renewable power, load, price of energy and various electric vehicle parameters based on adaptive robust programming.

The remainder of the paper is structured as follows: In Part 2, the nonlinear deterministic model and linear approximation formulation for SDN energy management including IESs are presented. Then, in Section "ARO-based uncertainty modeling", the formulation of different uncertainties according to ARO is presented. In Part 4, simulation finding obtained from various cases are reported to evaluate the feasibility of the suggested scheme. General conclusions are presented in Part 5.

The deterministic multi-objective energy management of SDN with IESs

Non-linear formulation

In this part, a deterministic nonlinear formulation of SDN energy management with renewable IESs is presented. Goal function is equal to minimizing the weighted sum of energy cost, voltage profile function, and emissions in SDN. It is constrained by the AC optimal power flow and grid voltage security limits, and the operation formulation of renewable IESs with stationary and mobile storages. In this model, only one scenario is considered, which includes the expected (average) amount of uncertainties. The details of the suggested design are presented in the following parts.

A) *Goal function*: This function for scheme is presented in Eq. (1). It minimizes the weighted sum of the normalized value of the SDN operating cost (EC)[21], the voltage deviation function (VD)[24], and the SDN emission (EM)[25]. The unit of each function is not the same. In other words, the aforementioned functions are expressed in terms of $, p.u., and ton, respectively. Therefore, the normalized value of different functions is given in Eq. (1). The normalized value of each function (F) is equal to the difference of F from its minimum value (F^{min}) divided by the changes range of the function. The changes range of a function is also equal to the difference of the maximum (F^{max}) and the minimum value of the function. The formulation of EC, VD, and EM is expressed in Eqs. (2–4), respectively. In Eq. (2), EC is equal to the purchased energy cost from the upstream network. The SDN is linked to the upstream network via a distribution substation. So, in Eq. (2), the active power of the distribution substation appears. EC per hour equals the product of the energy price and the energy passing through the distribution substation[21]. In Eq. (3), the voltage deviation function is expressed. VD equals the sum of the voltage deviations of the SDN buses from the desired value (here 1 p.u.) in all operating hours. Based on the minimization of VD, the voltage drop of the lines can be minimized, and consequently the power losses of the lines are also minimized. Therefore, the minimization of VD results in improving the SDN operating indicators. In addition, the minimization of VD obtains a safe voltage margin for different buses[24]. Therefore, it can also result in improving voltage security. The amount of environmental pollution in the network is calculated from Eq. (4). In this relation, CO2, SO2 and NOx emissions are considered. So that the total pollution coefficient in this relation is the sum of the CO2, SO2 and NOx emission coefficients. The energy received by SDN from the upstream network is accompanied by the aforementioned emissions, because this energy is received from centralized power plants[25]. These power plants generally consume fossil fuels. Therefore, in Eq. (4), the active power of distribution substations is used.

$$\min \quad OF = \varpi_{EC}\frac{EC - EC^{\min}}{EC^{\max} - EC^{\min}} + \varpi_{VD}\frac{VD - VD^{\min}}{VD^{\max} - VD^{\min}} + \varpi_{EM}\frac{EM - EM^{\min}}{EM^{\max} - EM^{\min}} \tag{1}$$

where:

$$EC = \sum_{b,h} \rho_{b,h} P_{DS\,b,h} \tag{2}$$

$$VD = \sum_{b,h} |V_{b,h} - 1| \tag{3}$$

$$EM = \sum_{b,h} \left(\zeta_{CO2} + \zeta_{SO2} + \zeta_{NOx}\right) P_{DS\,b,h} \tag{4}$$

In Eq. (1), ϖ_{EC}, ϖ_{VD} and ϖ_{EM} are the weight coefficients of EC, VD and EM in the objective function, respectively, and are in the range of [0, 1]. The sum of these weight coefficients must be unity[21]. Through changing the values of these coefficients, other values are calculated for the EC, VD and EM functions. Therefore, a large number of optimal points are available for the proposed design. The set of these optimal points represents the Pareto front for the suggested design. In this paper, the fuzzy decision-making technique is used to determine an optimal point. This point represents the compromise solution between the EC, VD and EM functions. The steps of this technique are as Algorithm 1[21].

- *Step* 1: Determine the minimum (F^{min}) and maximum (F^{max}) value of each function (F) for solving the problem for three cases, respectively, including $\varpi_{EC} = 1$, $\varpi_{VD} = 1$ and $\varpi_{EM} = 1$,
- *Step* 2: Determine the random value for the coefficients ϖ_{EC}, ϖ_{VD} and ϖ_{EM},
- *Step* 3: Solve the problem for the second step,
- *Step* 4: Find the value of the linear fuzzy membership function (f) for the EC, VD and EM functions:
  If F is smaller (greater) than F^{min} (F^{max}), then f is 1 (zero).
  If the value of F is greater than F^{min} and less than F^{max}, the value of f is equal to $(F - F^{max})/(F^{min} - F^{max})$.
- *Step* 5: Minimize $f(\phi)$, i.e. $\phi = \min\{f(\text{EC}), f(\text{VD}), f(\text{EM})\}$,
- *Step* 6: If the number of members of the Pareto front is exhausted, step 7 is performed, otherwise steps 2 to 5 are repeated.
- *Step* 7: Determine the compromise point corresponding to the maximum ϕ.

Algorithm 1. The process of the fuzzy decision-making.

B) *SDN constraints*: In Eqs. (5–15), the AC optimal power flow model with voltage security formulation for SDN is expressed. The AC power flow model of the network corresponds to the constraints (5–10)[4]. These equations respectively express the active-reactive power equilibrium in SDN buses[26–29], the active and reactive power flowing through distribution lines of the SDN, the amplitude and angle values of voltage in the slack bus. The constraints of SDN operation are expressed in Eqs. (11–13) [14]. In Eq. (11), the boundary on the voltage amplitude of the buses is presented[30–33]. Its upper limit is for observing insulation damage due to high overvoltage. Its lower boundary prevents consumer interruption in the case of higher drops in voltage. In constraint (12), the limitation of the distribution substation capacity is presented. This constraint expresses the limitation of the maximum apparent power in the distribution substation. The capacity limitation of SDN's distribution lines is also shown in Eq. (13). The voltage security model is expressed in Eqs. (14) and (15) [21]. In this paper, the worst-case security index (WSI) is applied to investigate the voltage security status of the network. WSI is calculated from Eq. (14) for a weak bus in terms of voltage amplitude between [0, 1], where 0 (1) represents the voltage collapse (no-load operation) of the network. The WSI limit is expressed in Eq. (15). To establish a safe voltage margin in SDN buses, the minimum WSI value based on Eq. (15) is generally equal to 0.8[21].

There are various indicators for voltage security, most of which are used in ring networks and transmission networks. In radial distribution networks, the WSI model is generally used[21]. Because it examines the voltage security condition in the weakest bus (include low voltage) which is at the end of the feeder. So its process is simple and uncomplicated. If security of voltage is established in the weak bus, voltage security will also be established in the other buses[21].

$$P_{DS\,b,h} + P_{I\,b,h} - P_{C\,b,h} = \sum_j A_{b,j} P_{DL\,b,j,h} \quad \forall b, h \tag{5}$$

$$Q_{DS\,b,h} - Q_{C\,b,h} = \sum_j A_{b,j} Q_{DL\,b,j,h} \quad \forall b, h \tag{6}$$

$$P_{DL\,b,j,h} = \alpha_{DL\,b,j} (V_{b,h})^2 - V_{b,h} V_{j,h} (\alpha_{DL\,b,j} \cos(\sigma_{b,h} - \sigma_{j,h}) + \beta_{DL\,b,j} \sin(\sigma_{b,h} - \sigma_{j,h})) \quad \forall b, h \tag{7}$$

$$Q_{DL\,b,j,h} = -\beta_{DL\,b,j} (V_{b,h})^2 + V_{b,h} V_{j,h} (\beta_{DL\,b,j} \cos(\sigma_{b,h} - \sigma_{j,h}) - \alpha_{DL\,b,j} \sin(\sigma_{b,h} - \sigma_{j,h})) \quad \forall b, h \tag{8}$$

$$V_{b,h} = 1 \quad \forall b = 1, h \tag{9}$$

$$\sigma_{b,h} = 0 \quad \forall b = 1, h \tag{10}$$

$$V_b^{lo} \le V_{b,h} \le V_b^{up} \quad \forall b, h \tag{11}$$

$$\sqrt{(P_{DS\,b,h})^2 + (Q_{DS\,b,h})^2} \le S_{DS\,b}^{up} \quad \forall b, h \tag{12}$$

$$\sqrt{(P_{DL\,b,j,h})^2 + (Q_{DL\,b,j,h})^2} \le S_{DL\,b,j}^{up} \quad \forall b, j, h \tag{13}$$

$$\chi_{W\,h} = (V_{p-1,h})^4 - 4(V_{p-1,h})^2 (R_{p,p-1} P_{DL\,p,p-1,h} + X_{p,p-1} Q_{DL\,p,p-1,h}) - 4(X_{p,p-1} P_{DL\,p,p-1,h} - R_{p,p-1} Q_{DL\,p,p-1,h})^2 \quad \forall h \tag{14}$$

$$\chi_{W\,h} \ge 0.8 \quad \forall h \tag{15}$$

C) *IES constraints*: In Eqs. (16–37), the operating model of a renewable IES with hydrogen storage, BEVs, PHEVs, and FCEVs is expressed. In Eq. (16), the active power balance in the IES is presented. According to this equation, the active power of the IES from the SDN perspective (P_I) is the sum of the active power generated by renewable sources, fuel cell (FC) in HS, and BEVs and PHEVs in discharge mode minus the active power consumed by passive loads, electrolyzer (EL) in HS, and BEVs and PHEVs in charge mode. The hydrogen balance in the IES is proportional to Eq. (17). Hydrogen is assumed to be not consumed or produced in the SDN. Therefore, in the IES, the produced hydrogen should be equal to the consumed hydrogen. In other words, the total hydrogen

produced by EL in HS, hydrogen tank (HT) in discharge mode, and FCEVs in discharge mode should be equal to the total hydrogen consumed by FC in HS, HT and FCEVs in charge mode. In Eq. (18), the amount of active power generation by the renewable source is expressed[14]. This power per hour is the product of the source size and the source power production rate (φ_R)[34–36]. Equation (18) can be used for different types of RES, with the difference that their φ_R values are different[37,38]. The power production rate in PV is equal to the ratio of PV irradiation and maximum solar radiation. In BS, the power production rate is equal to the BS gas production divided by the maximum amount of this gas[21]. In WS, φ_R is a function of wind speed. So that if the wind speed drops (exceeds) the cut-in (cut-out) speed, φ_R is equal to zero. If the wind speed has a value between the cut-in and nominal speed, φ_R will follow the wind speed linearly. Finally, if the wind speed is between the nominal and cut-out speeds, φ_R is equal to 1[14]. The combined performance of BEVs and PHEVs is given by (19–22)[24,39]. Constraints (19) and (20) express the charge and discharge rate limits[40–42] of these types of EVs, respectively. Constraint (21) calculates the amount of energy storage in the battery of these EVs[43–45]. This energy at hour h is the sum of initial energy in the BEVs and PHEVs newly connected to the IES, the energy storage in the previous hour, and the energy received from the charging operation of these EVs minus the sum of the final energy of the BEVs and PHEVs exited from the IES, and the discharged energy of these EVs in discharging operation[46,47]. Finally, the limit of the stored energy in the battery of BEVs and PHEVs is given by (22)[38]. In each hour, different types of BEVs and PHEVs with different capacities are connected to the IES. Therefore, the charging and discharging rates (CR and DR), initial and final energy in the integrated model of the mentioned EVs change with time[48,49]. Based on this, the index h is used for the parameters κ_{CH}, κ_{DIS}, E_A and E_D[38]. The HS performance model is shown in Eqs. (23–30)[50]. HS includes EL, FC and HT. EL transforms electricity into hydrogen. FC then transforms hydrogen into electricity. HT stores hydrogen. In Eqs. (23) and (24), the mathematical model of EL operation is presented[50]. In constraint (23), the amount of hydrogen generation by EL is calculated, which is a coefficient of the active power of EL. The capacity limit (active power) of the electrolyzer is expressed in Eq. (24). The FC performance is proportional to constraints (25) and (26). The hydrogen consumption rate of FC is proportional to Eq. (25). The capacity limit of FC is modeled in constraint (26). The HT performance model is also proportional to Eqs. (27–30). So that the charge and discharge rate limits appear in the constraints (27) and (28). The stored hydrogen energy in the HT is proportional to Eq. (29) [50], and at hour h, is the sum of energy stored in the previous hour and the energy obtained from the HT charging mode minus the energy discharged due to the HT discharging operation. At $h=1$, the initial energy is used instead of the stored energy in the previous hour. The hydrogen energy limit in the HT is proportional to Eq. (30). Equations (31–34) express the FCEV performance model, which respectively express the charge and discharge rate limits of FCEVs, the stored energy in FCEVs, and the stored energy limit in FCEVs. The stored hydrogen energy in FCEVs for hour h is the sum of initial hydrogen energy of FCEVs newly-linked to the IES, the energy stored in the previous hour, and the energy obtained from the charging operation of these vehicles minus the sum of final hydrogen energy of FCEVs exiting the IES and the energy discharged in the discharging mode of FCEVs. In each hour, FCEVs with different capacities are connected to the IES. Therefore, υ_{CH}^{fcev}, υ_{DIS}^{fcev}, E_P and E_F change with time. Accordingly, the index h is used for these parameters.

$$P_{I\,b,h} = P_{R\,b,h} + \left(P_{FC\,b,h} - P_{EL\,b,h}\right) + \left(P_{DIS\,b,h} - P_{CH\,b,h}\right) - P_{C\,b,h} \quad \forall b, h \tag{16}$$

$$H_{EL\,b,h} + \left(H_{DIS\,b,h}^{hs} - H_{CH\,b,h}^{hs}\right) + \left(H_{DIS\,b,h}^{fcev} - H_{CH\,b,h}^{fcev}\right) - H_{FC\,b,h} = 0 \quad \forall b, h \tag{17}$$

$$P_{R\,b,h} = P_{R\,b}^{up}\varphi_{R\,b,h} \quad \forall b, h \tag{18}$$

$$0 \le P_{CH\,b,h} \le \kappa_{CR\,b,h} \quad \forall b, h \tag{19}$$

$$0 \le P_{DIS\,b,h} \le \kappa_{DR\,b,h} \quad \forall b, h \tag{20}$$

$$E_{EV\,b,h} = E_{A\,b,h} + E_{EV\,b,h-1} + \eta_{CH}P_{CH\,b,h} - \frac{1}{\eta_{DIS}}P_{DIS\,b,h} - E_{D\,b,h} \quad \forall b, h \tag{21}$$

$$E_{EV\,b,h} \ge 0 \quad \forall b, h \tag{22}$$

$$H_{EL\,b,h} = \eta_{EL}P_{EL\,b,h} \quad \forall b, h \tag{23}$$

$$0 \le P_{EL\,b,h} \le P_{EL\,b}^{up} \quad \forall b, h \tag{24}$$

$$H_{FC\,b,h} = \frac{1}{\eta_{FC}}P_{FC\,b,h} \quad \forall b, h \tag{25}$$

$$0 \le P_{FC\,b,h} \le P_{FC\,b}^{up} \quad \forall b, h \tag{26}$$

$$0 \le H_{CH\,b,h}^{hs} \le \upsilon_{CR\,b}^{hs} \quad \forall b, h \tag{27}$$

$$0 \le H_{DIS\,b,h}^{hs} \le \upsilon_{DR\,b}^{hs} \quad \forall b, h \tag{28}$$

$$E_{HT\,b,h} = z_h E_{HT\,b}^{in} + \left(z_h - 1\right) E_{HT\,b,h-1} + H_{CH\,b,h}^{hs} - H_{DIS\,b,h}^{hs} \quad \forall b, h, z = 1(0) | h = 1(\neq 1) \tag{29}$$

$$E_{HT\,b}^{lo} \le E_{HT\,b,h} \le E_{HT\,b}^{up} \quad \forall b, h \tag{30}$$

$$0 \le H_{CH\,b,h}^{fcev} \le \upsilon_{CR\,b,h}^{fcev} \quad \forall b, h \tag{31}$$

$$0 \leq H_{DIS\,b,h}^{fcev} \leq \upsilon_{DR\,b,h}^{fcev} \quad \forall b, h \tag{32}$$

$$E_{ET\,b,h} = E_{I\,b,h} + E_{ET\,b,h-1} + H_{CH\,b,h}^{fcev} - H_{DIS\,b,h}^{fcev} - E_{F\,b,h} \quad \forall b, h \tag{33}$$

$$E_{ET\,b,h} \geq 0 \quad \forall b, h \tag{34}$$

Linear approximation model

Formulation (1–34) is nonlinear and non-convex due to the AC power flow model[51]. However, according to Section "Research contributions", the main goal of this paper is robust modeling of the suggested design according to ARO. This technique requires a linear model of the presented design[24]. In addition, linearization of the problem allows to reach the unique optimal solution by different solution algorithms. This is not the case in the nonlinear model. Also, the scheme is one of the power system operation problems that need short computing time. The linear model has short computing time than that of the nonlinear model[24]. Considering the above, the linear approximation model for the presented design is presented here.

To linearize the power flow equations, especially Eqs. (7) and (8), it should be noted that the voltage angle deviation between the sending and receiving ends of the line in the distribution network is less than 6°[24]. Accordingly, $\cos(\sigma_b - \sigma_j)$ is roughly 1, and $\sin(\sigma_b - \sigma_j)$ is approximately equal to $(\sigma_b - \sigma_j)$. In addition, in the optimal operation of SDN, the voltage amplitude is expected to be between 0.95 and 1.05 p.u. In other words, in optimal conditions, the voltage deviation ($\Delta V = V - 1$) is between -0.05 p.u. and 0.05 p.u. This amount is much less than the desired value of the voltage amplitude (1 p.u.). Therefore, the expression $1 + \Delta V$ replaces the voltage amplitude (V). Next, by ignoring the insignificant values of $\Delta V_b \Delta V_j$, ΔV^2 and $\Delta V(\sigma_b - \sigma_j)$, the linear approximation model of Eqs. (7) and (8) can be presented as Eqs. (35) and (36), respectively[24]:

$$P_{DL\,b,j,h} = \alpha_{DL\,b,j}\left(\Delta V_{b,h} - \Delta V_{j,h}\right) - \beta_{DL\,b,j}\left(\sigma_{b,h} - \sigma_{j,h}\right) \quad \forall b, j, h \tag{35}$$

$$Q_{DL\,b,j,h} = -\beta_{DL\,b,j}\left(\Delta V_{b,h} - \Delta V_{j,h}\right) - \alpha_{DL\,b,j}\left(\sigma_{b,h} - \sigma_{j,h}\right) \quad \forall b, j, h \tag{36}$$

Relations (9) and (11) are also expressed as constraints (37) and (38), respectively:

$$\Delta V_{b,h} = 0 \quad \forall b = 1, h \tag{37}$$

$$V_b^{lo} - 1 \leq \Delta V_{b,h} \leq V_b^{up} - 1 \quad \forall b, h \tag{38}$$

ΔV has two components, positive (ΔV^+) and negative (ΔV^-). ΔV^+ and ΔV^- are positive continuous variables based on constraints (39) and (40) which include positive and negative values of ΔV, respectively. In other words, ΔV is equal to ΔV^+ minus ΔV^- according to constraint (41). The expression $|\Delta V|$ will be equal to the sum of ΔV^+ and ΔV^-. In this case, the linearized model of the objective function (3) is given by (42):

$$0 \leq \Delta V_{b,h}^{+} \leq V_b^{up} - 1 \quad \forall b, h \tag{39}$$

$$0 \leq \Delta V_{b,h}^{-} \leq 1 - V_b^{lo} \quad \forall b, h \tag{40}$$

$$\Delta V_{b,h} = \Delta V_{b,h}^{+} - \Delta V_{b,h}^{-} \quad \forall b, h \tag{41}$$

$$VD = \sum_{b,h}\left(\Delta V_{b,h}^{+} + \Delta V_{b,h}^{-}\right) \tag{42}$$

The capacity of the substation and distribution lines in Eqs. (12) and (13) is a circular plane, i.e. $\sqrt{P^2 + Q^2} \leq S$. This equation can be linear if the circular plane is approximated by a regular polygon[24]. The equation of the tangent line to the boundary of the circle ($\sqrt{P^2 + Q^2} = S$) at an angle $k \times \Delta\theta$ is expressed as $P \times \cos(k \times \Delta\theta) + Q \times \sin(k \times \Delta\theta) = S$. k represents the side in the regular polygon. $\Delta\theta$ represents the deviation of the angle, and it is equal to $360/n_k$. n_k represents the number of sides in the regular polygon. The square plane corresponding to the aforementioned equation is $P \times \cos(k \times \Delta\theta) + Q \times \sin(k \times \Delta\theta) \leq S$. Repeating this relation for all values of k, i.e. $\{1, 2, \ldots, n_k\}$, will give a regular polygon whose sides are tangent to the boundary of the circle[24]. Accordingly, the linear approximation model of constraints (12) and (13) will be in the form of relations (43) and (44), respectively:

$$P_{DS\,b,h}\cos\left(k \times \Delta\theta\right) + Q_{DS\,b,h}\sin\left(k \times \Delta\theta\right) \leq S_{DS\,b}^{up} \quad \forall b, h, k \tag{43}$$

$$P_{DL\,b,j,h}\cos\left(k \times \Delta\theta\right) + Q_{DL\,b,j,h}\sin\left(k \times \Delta\theta\right) \leq S_{DL\,b,j}^{up} \quad \forall b, j, h, k \tag{44}$$

The WSI model in Eq. (14) is nonlinear. According to[21], the first term of this equation has a value greater than 0.8 p.u., the second term has a value less than 0.03 p.u., and the third term has a value less than 0.005 p.u. Therefore, it can be stated that WSI is approximately equal to the first term, i.e. $\chi_{W\,h} = \left(V_{p-1,h}\right)^4$. This equation is also modeled approximately as Eq. (45) based on $V = 1 + \Delta V$.

$$\chi_{W\,h} = 1 + 4\Delta V_{p-1,h} \quad \forall h \tag{45}$$

Of course, to ensure the establishment of a safe voltage margin in SDN based on the proposed linear approximation model, the minimum WSI value in Eq. (15) is increased to 0.83. In other words, Eq. (46) replaces Eq. (15):

$$\chi_{W\,h} \geq 0.83 \quad \forall h \tag{46}$$

Finally, the linear approximation model of the proposed scheme is formulated as (47) and (48):

$$\text{Eq. (1)} \tag{47}$$

Subject to:

$$\text{Eqs. } (2)\,,\ (4)-(6)\,,\ (10)\,,\ (16)-(46) \tag{48}$$

ARO-based uncertainty modeling

In the formulation (47, 48), parameters such as active and reactive power of the load, P_C and Q_C, renewable power rate, φ_R, energy price, ρ, CR and DR of BEVs and PHEVs, κ_{CR} and κ_{DR}, initial and consumed energy required by these vehicles, E_A and E_D, CR and DR of FCEVs, υ_{CH}^{fcev} and υ_{DIS}^{fcev}, initial and required consumed energy by FCEVs, E_I and E_F, have uncertainties. The proposed scheme has 12 uncertainty parameters. The uncertainty parameter matrix ($\overline{u}$) is defined as Eq. (49). This matrix has n rows and m columns. n is equal to the number of network buses (n_B), and m is the number of operating hours (n_T) multiplied by 12. In this paper, in order to reach a robust solution to the uncertainty prediction error, the ARO technique is used to model the aforementioned uncertainties. In this method, there is a scenario that includes the worst uncertainty values proportional to the objective function[52]. This scenario is known as the worst case scenario. Therefore, in ARO, uncertainties are known as uncertainty variables[52]. In other words, in ARO, the uncertainty variables matrix ($\hat{u}$) is used as Eq. (50). In the ARO model, the value of the uncertainty set (U) is determined. The elements of row b of this set can be calculated from Eq. (51) [52]. In this equation, $\tilde{u}$ is known as the uncertainty deviation. BU represents the uncertainty budget, which has a value between zero and one. Its value of zero means that the amount of the uncertainty variables is equal to their predicted value. In other words, in this situation, the deterministic model is considered for the problem. If BU has a value of 1, this means that all the parameters in Eq. (50) have uncertainty. The value $BU>0$ corresponds to the robust model. $[\overline{u}_i-\tilde{u}_i, \overline{u}_i+\tilde{u}_i]$ represents the changes range of $\hat{u}$. Finally, the set of uncertainties (U) can be defined as $\bigcup_b U_b$ [52].

$$\overline{u} = \begin{bmatrix} \rho & P_C & Q_C & \varphi_R & \kappa_{CR} & \kappa_{DR} & E_A & E_D & \upsilon_{CR}^{fcev} & \upsilon_{DR}^{fcev} & E_I & E_F \end{bmatrix}_{n\times m} \tag{49}$$

$$\hat{u} = \begin{bmatrix} \hat{\rho} & \hat{P}_C & \hat{Q}_C & \hat{\varphi}_R & \hat{\kappa}_{CR} & \hat{\kappa}_{DR} & \hat{E}_A & \hat{E}_D & \hat{\upsilon}_{CR}^{fcev} & \hat{\upsilon}_{DR}^{fcev} & \hat{E}_I & \hat{E}_F \end{bmatrix}_{n\times m} \tag{50}$$

$$U_b = \left\{ \hat{u}_b \in R^m : \frac{1}{m}\sum_{i=1}^{m} \frac{|\hat{u}_{b,i}-\overline{u}_{b,i}|}{\tilde{u}_{b,i}} \leq BU_b,\ \ \hat{u}_{b,i} \in [\overline{u}_i-\tilde{u}_i, \overline{u}_i+\tilde{u}_i] \ \ \forall i \in \{1,2,...,m\} \right\} \quad \forall b \tag{51}$$

Formulation (47, 48) is consistent with model (52–54). In formulation (52–54), y represents the vector of the main variables of the problem including voltage, active and reactive power, energy and hydrogen. f represents the matrix of coefficients in the objective function. Equations (53, 54) are equivalent to constraints without / with the uncertainty parameter. It is noteworthy that the constraint $a=b$ in problem (47, 48) can be defined as the sum of $a\leq b$ and $a\leq -b$.

$$\min \quad f \times y \tag{52}$$

Subject to:

$$g \times y \leq c \tag{53}$$

$$s \times y \leq \overline{u} \tag{54}$$

In the ARO technique, the worst-case scenario and the optimal point in this scenario are determined simultaneously[24,52]. Therefore, the ARO model for Eqs. (52–54) is formulated as Eq. (55). Since the objective function (1) has a "min" expression, the "max" expression should be used to determine the worst value of uncertainties in the worst-case scenario. The "min" expression in Eq. (55) is used to extract the optimal solution in the worst-case scenario. The variable y is determined from the set Ξ, which is based on Eq. (56) containing the constraints of the proposed problem in exchange for the uncertainty variable replacing the uncertainty parameter. In this equation, μ represents the dual variable of the problem constraints. The uncertainty variable is also obtained from the set U, which is formulated as Eq. (57) [52].

$$\min_{y\in\Xi} \max_{\hat{u}\in U} \quad f \times y(\hat{u}) \tag{55}$$

where:

$$\Xi \triangleq \{y(\hat{u}) \ \mid \ g \times y(\hat{u}) \leq c : \mu_1,\ \ s \times y(\hat{u}) \leq \hat{u} : \mu_2\} \tag{56}$$

$$U \triangleq \bigcup_{b} U_b \tag{57}$$

In order to solve relations (55–57) simply, the objective function (55) is transformed into model (58). To solve this problem, the dual problem of $\min_{y\in\Xi} \quad f \times y(\hat{u})$ is first extracted. In the dual problem, the term "min" is transformed into "max". Therefore, the ARO model of the presented design will include the term "max-max", which is equivalent to "max"[52].

$$\max_{\hat{u}\in U} \min_{y\in\Xi} \quad f \times y(\hat{u}) \tag{58}$$

Finally, the final ARO model for problem (52–54) will be in the form of relations (59–61). In relation (59), the expression $\max_{\mu_1,\mu_2} \quad c \times \mu_1 + \hat{u} \times \mu_2$ is the dual of the objective function $\min_{y\in\Xi} \quad f \times y(\hat{u})$[52]. $\max_{\hat{u},\mu_1,\mu_2} \quad c \times \mu_1 + \hat{u} \times \mu_2$ is also equivalent to relation (58). Relation (60) represents the dual constraint model of the original problem or relation (56). In this relation, if the variable y has only positive/negative values, then the expression $\leq / \geq$ is used. If y includes both positive and negative values, the expression "$=$" is used[24]. The worst-case scenario constraint is also equal to relation (61) [52].

$$\max_{\hat{u},\mu_1,\mu_2} \quad c \times \mu_1 + \hat{u} \times \mu_2 \tag{59}$$

Subject to:

$$g \times \mu_1 + s \times \mu_2 \leq / = / \geq f \tag{60}$$

$$U \triangleq \bigcup_{b} U_b \tag{61}$$

In this paper, the standard model of the proposed scheme, (47, 48), was first presented in the formulation (52–54). Then, based on the ARO technique, the robust model of the proposed scheme is similar to the model (59–61). According to this model, the robust model for a problem is equal to the duality model of that problem with the addition of the uncertainty set constraint, i.e. (61). In addition, since the robust model has a scenario, its computational time is expected to be very low compared to stochastic optimization. This is confirmed in[24,52]. Since the robust model obtains the worst-case scenario for uncertainties[53], the extracted optimal solution is robust against the uncertainty prediction error[24]. One of the disadvantages of the ARO model is that it does not consider the probability of the worst-case scenario occurring. Therefore, the worst-case scenario may occur with a very low probability.

Simulation results

Case study

The proposed scheme with the formulation in Sections "The deterministic multi-objective energy management of SDN with IESs" and "ARO-based uncertainty modeling" can be applied to different distribution networks, different data of resources and storage devices. The scheme with the model (47, 48) based on robust modeling of uncertainties in Section "ARO-based uncertainty modeling" is tested on the IEEE 69-bus SDN[54] as shown in Fig. 2. The base power is 1 MVA, while the base voltage is 12.66 kV. Bus 1 is known as the reference bus. The amplitude and phase angle of the voltage in this bus are 1 p.u. and zero radian, respectively. The minimum and maximum allowable values of the voltage magnitude in each bus are 0.95 p.u. and 1.05 p.u.[55–59]. In[54], the characteristics of the distribution lines including resistance, reactance, conductance, susceptance and maximum apparent power passing through it are presented. The capacity of the distribution substation is also 4.6 MVA according to[54]. The peak active and reactive power values for the load in each bus are extracted from[54]. The load amount in other hours is obtained by multiplying the peak load and the load factor. The load factor value at different hours is proportional to Fig. 3 [14]. The energy price for a 24 h operating horizon has three values that change in a stepwise manner. In other words, according to[24], it is 16 (30) $/MWh in the interval 1:00–7:00 (17:00–22:00). In other hours, it has a value of 24 $/MWh. The sum of the emission coefficients for receiving energy from the upstream network is 918.2 ton/MW[25]. The aforementioned network has 8 IESs. The location of these IES is shown in Fig. 2. The characteristics of renewable sources, electric vehicles and hydrogen storage in each IES are proportional to Table 2. In this table, the capacity (maximum generated active power) of each renewable source is presented. The power generation rate of each source, namely wind system, solar panel, and bio-waste unit, for different operating hours is shown in Fig. 3 [14,21]. In Table 2, the total number of BEVs, PHEVs and FCEVs that can be powered by the IES is given. Their number per hour is obtained by multiplying the total number of vehicles and the EV penetration rate. The EV penetration rate per hour can be obtained from Fig. 3 [24,39]. In this curve, it is assumed that EVs enter the IES after their last trip during the day and then perform charging and discharging operations. They also randomly exit the IES from 4:00 to 9:00[24]. The specifications of BEVs and PHEVs including battery capacity, battery CR and DR, initial and consumption energy required by the vehicles are presented in[24,39]. The charging and discharging efficiency of these EVs is 93% and 92%, respectively[24,26]. For FCEVs, the initial energy is 20% of the hydrogen tank capacity in the FCEV. It is assumed that each FCEV tends to have its hydrogen tank 100% full after leaving the IES. The charging (discharging) rate of each FCEV is the ratio of the hydrogen tank capacity and the charging (discharging) time. The charging and discharging time for each FCEV is assumed to be 2 h. It is assumed that their hydrogen tank capacity in

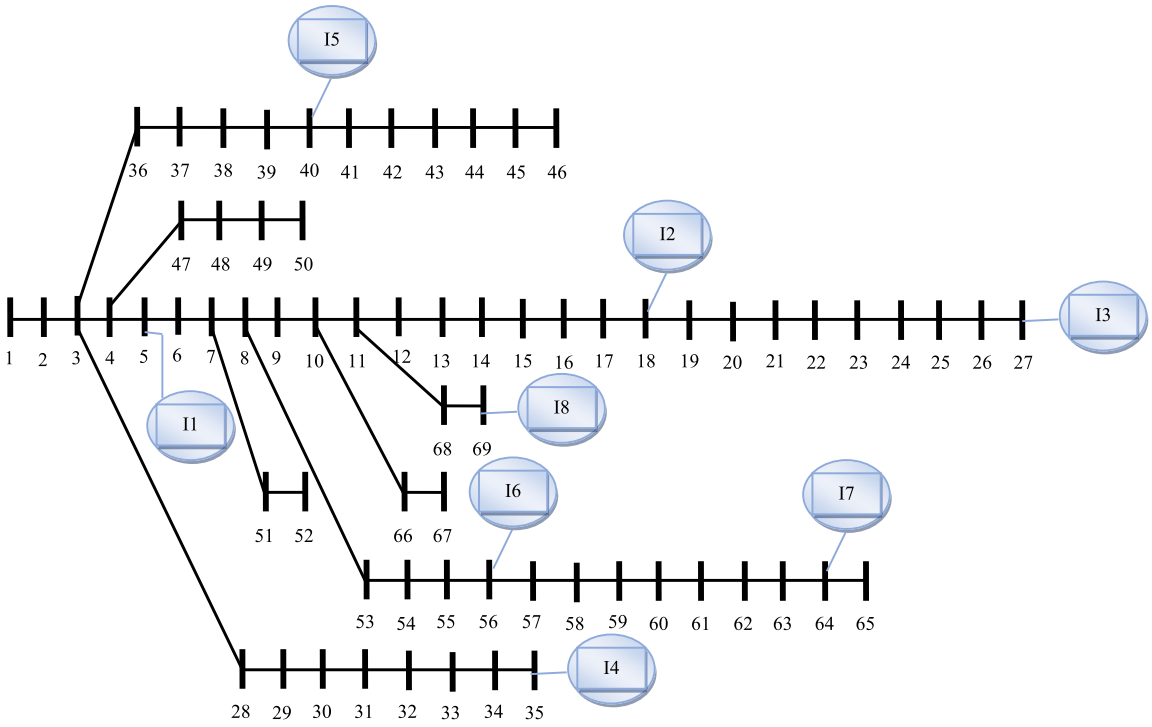


Fig. 2. IEEE 69-bus radial distribution network[54] considering IESs.

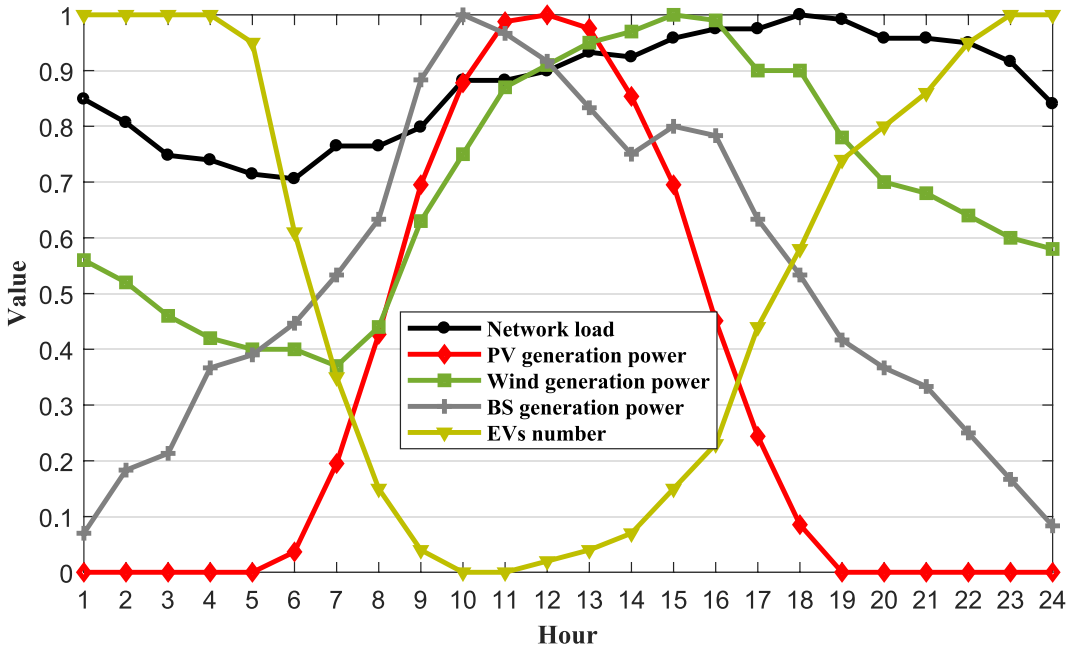


Fig. 3. Daily rate value curve of network load[14], RES generation power[14,21] and EVs number[24,39].

MWh is the same as the battery capacity data in BEVs. The charging and discharging rate of the aggregate BEVs per hour is the sum of the CR and DR of the BEVs connected to the IES in that hour. The initial energy (required consumption energy) of the aggregate BEVs at any moment is the sum of the initial energy (required consumption energy) of the newly-linked BEVs to (exited from) the IES in that hour. This is also true for PHEVs and FCEVs.

IES		I1	I2	I3	I4	I5	I6	I7	I8
P_R^{up} (MW) for	WS	1.10	0	0	0	0.7	0	0	0.8
	PV	0	0	0.30	0	0	0	0.30	0
	BS	0	0.60	0	0.60	0	0.5	0	0
Peak value of P_C (MW)		0.20	0.20	0.03	0.10	0.10	0.10	0.03	0.20
Number of	BEVs	20	20	10	0	20	0	5	20
	PHEVs	30	30	15	0	30	0	10	30
	FCEVs	15	0	0	30	0	20	5	15
P_{EL}^{up} (MW)		1.10	0	0	0.60	0.70	0.50	0.30	0.8
η_{EL} (%)		75	0	0	75	75	75	75	75
P_{FC}^{up} (MW)		1.10	0	0	0.60	0.70	0.50	0.30	0.8
η_{FC} (%)		51	0	0	51	51	51	51	51
v_{CR}^{hs} (MW)		0.80	0	0	0.50	0.60	0.45	0.25	0.7
v_{DR}^{hs} (MW)		0.80	0	0	0.50	0.60	0.45	0.25	0.7
E_{HT}^{in} (MWh)		0.30	0	0	0.20	0.20	0.2	0.10	0.3
E_{HT}^{lo} (MWh)		0.30	0	0	0.20	0.20	0.2	0.10	0.3
E_{HT}^{up} (MWh)		3	0	0	2	2	2	1	3

Table 2. Data of IESs.

Results and discussion

Numerical results for the proposed scheme corresponding to the information in the previous part are extracted from the GAMS platform. The robust model of the suggested design according to (59–61) includes linear constraints and a quadratic objective function. The problem can be solved in the GAMS software by the CPLEX algorithm[60]. In the linearized model, a circular plane is approximated by a regular 90-sided shape. According to[24], in the proposed linearized model for the optimal power flow problem in the distribution network, the computational error for power and voltage is around 1.3% and 0.4%, respectively. The uncertainty deviation is assumed to be the multiplication of the budget of uncertainty (BU) and the uncertainty radius (R).

(A) *Review of the economic, environmental and technical objectives of the DSO*: In Fig. 3, the Pareto front for the proposed scheme in the deterministic model (R = BU = 0) is presented. In this figure, the values of the objective functions EC, VD and EM are reported for values of zero, 0.25, 0.33, 0.5, 0.75 and 1 for the weighting factors ϖ_{EC}, ϖ_{VD} and ϖ_{EM}. The minimum and maximum value of EC based on this figure are equal to 1030 $ and 1870 $, respectively. These values for VD (EM) are equal to 21.3 p.u. (21,200 tons) and 48.2 p.u. (43,500 tons), respectively. The range of variations of each function is equal to the maximum value minus the minimum value of the function. Therefore, the extension of variations of functions EC, VD and EM is 840 $, 26.9 p.u. and 22,300 tons, respectively. In addition, it is observed from Fig. 3 that the direction of the changes of the mentioned functions is not the same. For example, the reduction of energy cost is proportional to the increase of VD. Because in the minimization of EC, local sources or IES inject high active power into the distribution network. In this situation, it is possible to create overvoltage in the buses, which leads to an increase in the voltage deviation function or VD. In Fig. 4, the compromise point between the EC, VD and EM functions obtained from the fuzzy decision-making algorithm for different values of BU and R in the robust model is reported. To derive this figure, the step of changes of the weight coefficients ton equal to 0.02 was used in the simulation. In the deterministic model (R = BU = 0), the amount of the EC, VD and EM functions at the compromise point according to Fig. 4 is equal to 1230 $, 27.2 p.u. and 26,500 ton, respectively. These values are close to the minimum value of EC, VD and EM, respectively. So that the fuzzy decision-making algorithm finds a value for EC at the compromise point that is about 23.8% ((1230–1030) ÷ 840) further from its minimum. This value is equal to 21.9% and 23.3% for VD and EM, respectively.

According to Fig. 4, the values of EC, VD and EM functions in the robust model are higher than in the deterministic model. So that by increasing the value of BU to 0.5, the higher R raises the mentioned functions compared to the deterministic model. By increasing BU to 1, increasing R to 0.3 increases the EC, VD and EM functions. However, for R > 0.3, the values of the mentioned functions do not change compared to the condition R = 0.3. In other words, the proposed problem has a robust solution for a maximum uncertainty radius equal to 0.3. This means that the proposed scheme is robust for a maximum prediction error of 30% for load, renewable power, energy price and EV parameters uncertainties. In Table 3, the amount of uncertainty variables in the worst case scenario is reported for different values of the uncertainty budget and uncertainty radius. According to this table, the increase in R and BU, in general, the increase in uncertainty deviation, causes to increase the energy price, active and reactive power of the load, and the energy consumption required by EVs than the deterministic model. However, in this situation, the power generation rate of RES, the initial energy of EVs, and the charging and discharging rate of EVs decrease. Therefore, energy consumption and energy price increase in the worst-case scenario than the scenario corresponding to the deterministic model, while the capacity of RES and EVs to produce energy decreases. In the case of EVs, the increase in R and BU causes to increase the energy

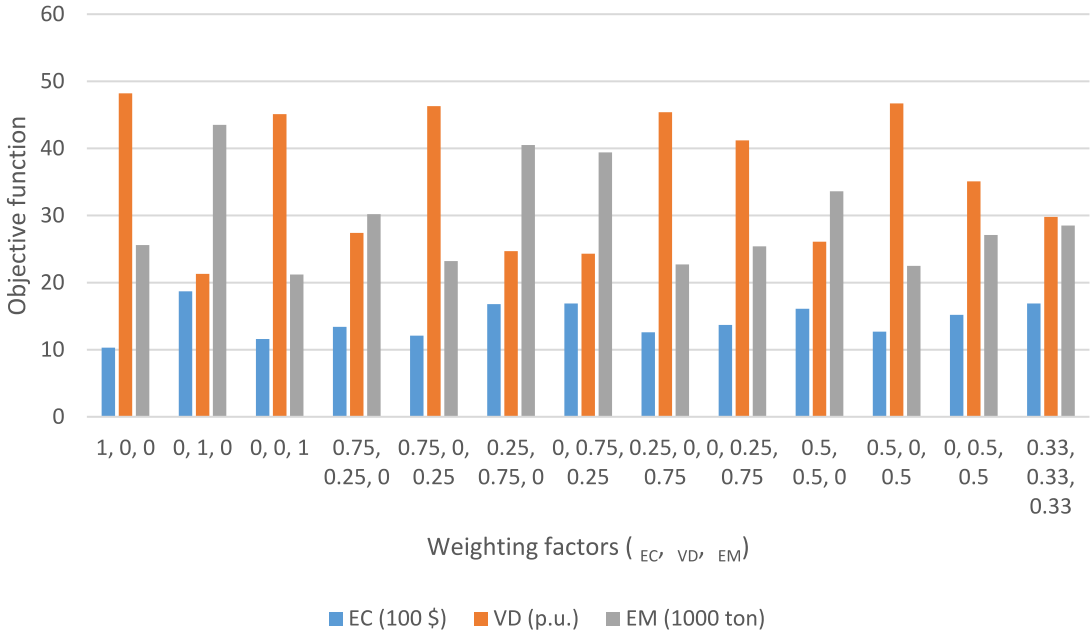


Fig. 4. Pareto front of the suggested design for the deterministic model ($R = BU = 0$).

R, BU	0, 0	0.1, 0.5	0.2, 0.5	0.1, 1	0.2, 1
$\sum_{b,h} \hat{\rho}_{b,h}$	556	583.8	611.6	611.6	667.2
$\sum_{b,h} \hat{P}_{C,b,h}$	74.3	78	81.7	81.7	89.1
$\sum_{b,h} \hat{Q}_{C,b,h}$	46.4	48.7	51	51	55.6
$\sum_{b,h} \hat{\varphi}_{R,b,h}$	36.5	34.7	32.9	32.9	29.3
$\sum_{b,h} \hat{\kappa}_{CR,b,h}$	9.4	8.9	8.4	8.4	7.5
$\sum_{b,h} \hat{\kappa}_{DR,b,h}$	9.4	8.9	8.4	8.4	7.5
$\sum_{b,h} \hat{E}_{A,b,h}$	0.58	0.55	0.52	0.52	0.46
$\sum_{b,h} \hat{E}_{D,b,h}$	2.9	3.05	3.2	3.2	3.5
$\sum_{b,h} \hat{v}^{fcev}_{CR,b,h,}$	3.8	3.6	3.4	3.4	3
$\sum_{b,h} \hat{v}^{fcev}_{DR,b,h}$	3.8	3.6	3.4	3.4	3
$\sum_{b,h} \hat{E}_{I,b,h}$	0.24	0.23	0.22	0.22	0.20
$\sum_{b,h} \hat{E}_{F,b,h}$	1.2	1.26	1.32	1.32	1.44

Table 3. Uncertainty parameter values in the worst-case scenario for different *Rs* and *BUs*.

requested by EVs (final energy—initial energy) from IESs. In addition, the CR and DR of EVs have decreased in this situation. Therefore, EVs need more time to charge themselves than in the deterministic model.

(B) *Analyzing the performance of IESs in the distribution network*: In Fig. 5, the daily active power curve of RESs, BEVs and PHEVs, fuel cell and electrolyzer along with the daily hydrogen power curve of FCEVs and

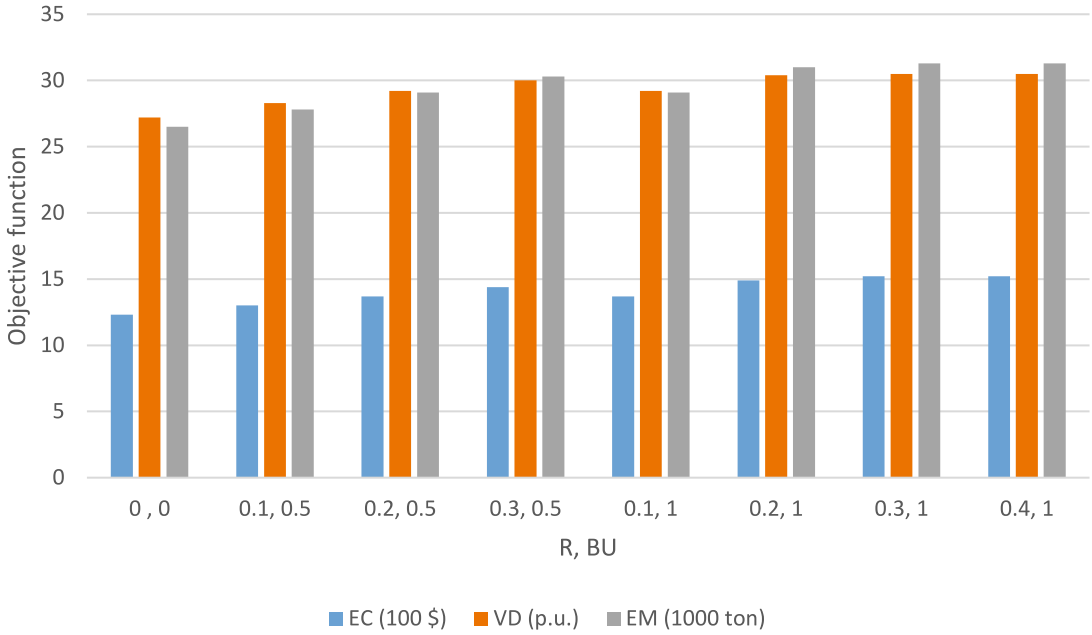


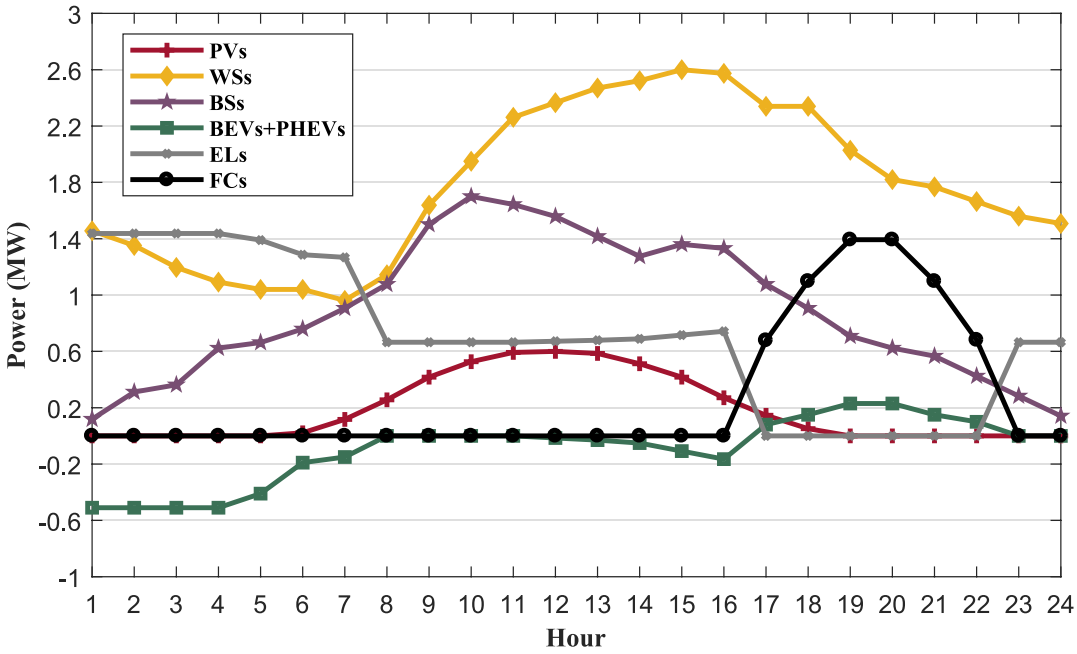
Fig. 5. Robust compromise solution of scheme considering various uncertainty levels and budgets.

hydrogen tank is plotted for the deterministic model. According to Fig. 5a, the daily active power curve of RESs is the same as the daily generation power rate curve of these sources in Fig. 6, they are only numerically different. This can be proven in Eq. (18). BEVs and PHEVs perform charging operations during off-peak hours when the energy price is the cheapest until they provide the required consumption energy for their travel. They perform charging operations again during the hours of 12:00–16:00 when the energy price is 24 $/MWh. During these hours, RESs also inject high power into the network. During the hours of 17:00–22:00, when the energy price is the expensive, the mentioned vehicles are in discharge mode and inject energy into the IESs. This performance of the mentioned electric vehicles corresponds to the minimization of energy cost, grid pollution and voltage profile function. The electrolyzers are active during the off- and mid-load hours, 1:00–16:00 and 23:00–24:00, and receive power from the IESs. However, the fuel cells are active during the peak hours, 17:00–22:00, and inject energy into the IESs. The electrolyzers were either active during the hours of cheap energy price, or they are active during the hours when the renewable resources are producing high power. The fuel cells are also active during the hours when the energy price is high. According to Fig. 5b, FCEVs, like other electric vehicles, charge during 1:00–7:00 to meet their required consumption energy. During these hours, the energy price is low, so the electrolyzer receives high power from the IES during these hours. FCEVs are also in charge mode during 12:00–16:00. They are in discharge mode during 17:00–22:00. During off-peak and mid-peak hours, the difference between the hydrogen produced by the electrolyzers and the hydrogen consumed by the FCEVs is injected into the hydrogen tank. However, during 17:00–22:00, the FCEVs and the hydrogen tank are in discharge mode, and they supply hydrogen to the fuel cell. This operation minimizes the operating cost, voltage profile function, and grid emissions based on Eq. (1). The daily active power curve of IESs for different values of uncertainty radius and uncertainty budget is plotted in Fig. 7. According to this figure, IESs have been able to be in energy generating mode for most of the hours. They have low power during off-peak hours, 1:00–7:00, because according to Fig. 5, during these hours, electric vehicles and electrolyzers received high energy from IESs. With increasing R and BU, the daily active power curve of IESs shifts downwards compared to the deterministic model. This is because in the worst-case scenario, the energy consumption of passive loads and EVs increases, and the power generation capacity of RESs and EVs decreases.

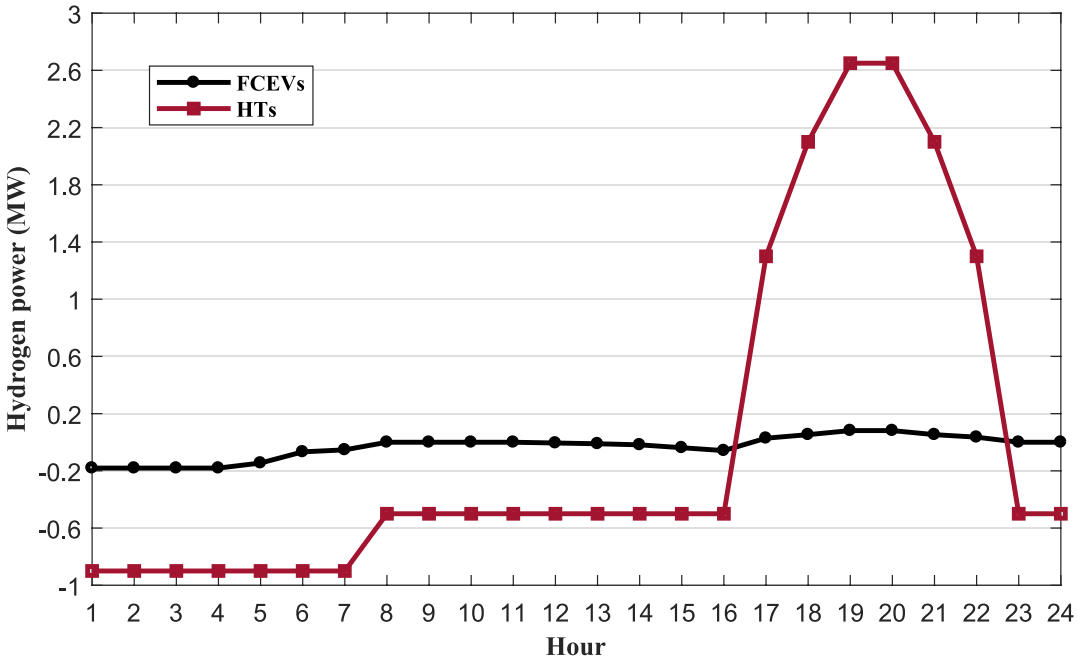
(C) *Investigating the effects of IES energy management on the technical, economic, and environmental status of the distribution network*: This section includes the following study items:

- Case I: Load flow study (network without IESs)
- Case II: Proposed scheme considering only RESs in IESs
- Case III: Case II + BEVs and PHEVs
- Case IV: Case II + HSs and FCEVs
- Case V: Case III + HSs and FCEVs

The values of the grid operating cost (EC), grid emission (EM), weak bus (p), minimum WSI, grid energy losses, maximum voltage drop and overvoltage, and peak load carrying capability (PLCC) for the aforementioned cases and BU = 1 are reported in Table 4. For each case, the values of the aforementioned indicators for the deterministic model (R = 0) and the robust model with maximum uncertainty radius (R = 0.3) are presented in this table. PLCC means how much peak load can supply by the distribution network. The worst case for the economic (EC), environmental (EM), operation (energy losses, maximum voltage drop, and PLCC), and



(a)



(b)

Fig. 6. Daily curve of, (**a**) active power of RESs, BEVs + PHEVs, FCs and ELs, (**b**) hydrogen power of HTs and FCEVs for the deterministic model ($R = BU = 0$).

voltage security (WSI) indicators was obtained for the load flow study (Case I). The presence of RESs alone in the IESs (Case II) was able to improve the EC, EM, WSI, energy losses, maximum voltage drop, and PLCC compared to Case I. However, this case is obtained for 0.023 p.u. increase in the maximum overvoltage. With the addition of BEVs and PHEVs to Case II (Case III), the status of various indicators is better than Case I. But compared to Case II, the EC, EM and network energy losses increase. This is because BEVs and PHEVs increase the energy demand of the distribution network from the upstream network in a 24 h horizon. But the energy management of these vehicles has caused Case III to have a more favorable condition for maximum voltage drop and overvoltage, PLCC and WSI than Case II. The addition of hydrogen storage and FCEVs to Case II (Case IV) causes the status of economic, environmental, operation and voltage security indicators to be better than Cases

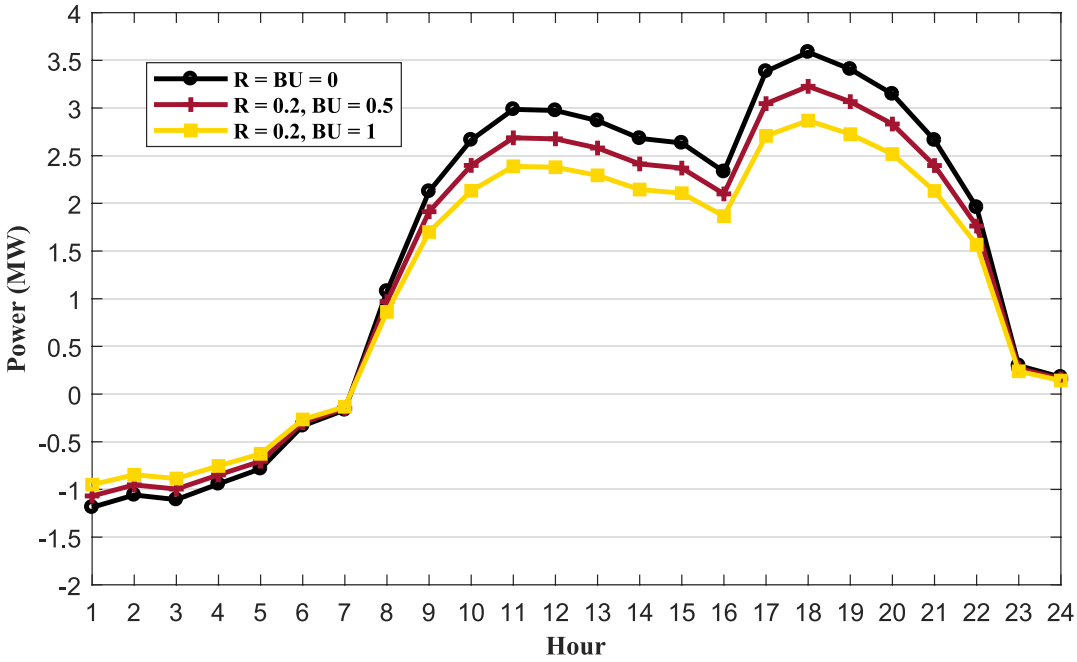


Fig. 7. Daily active power curve of IESs for the different values of *R* and *BU*.

Case	I		II		III		IV		V	
R	**0**	**0.3**	**0**	**0.3**	**0**	**0.3**	**0**	**0.3**	**0**	**0.3**
Network operation cost ($)	21.4	27.5	13.5	16.4	14.1	16.9	11.9	14.8	12.3	15.2
Network pollution (1000 ton)	69.7	90.8	28.9	33.8	29.6	34.4	25.8	30.6	26.5	31.3
Poor bus	65	65	65	65	65	65	65	65	65	65
Min value of WSI (p.u.)	0.73	0.68	0.86	0.83	0.90	0.88	0.89	0.86	0.92	0.90
Network energy loss (MWh)	3.12	3.94	2.32	3.04	2.43	3.15	2.24	2.92	1.78	2.42
Maximum voltage drop (p.u.)	0.092	0.128	0.046	0.049	0.041	0.043	0.043	0.045	0.039	0.041
Maximum over-voltage (p.u.)	0	0	0.023	0.020	0.015	0.013	0.017	0.015	0.012	0.010
PLCC (MW)	3.8	3.8	4.8	4.1	5.4	4.8	5.1	4.5	5.8	5.2

Table 4. Value of operation, economic, environmental and voltage security indices for different cases and *R* considering $BU = 1$.

I and II. In the comparison between Cases III and IV, energy losses, operation costs and network emissions are more favorable in Case IV, but other indicators are better in Case III. Therefore, the optimal performance of BEVs and PHEVs achieves more optimal and favorable conditions for WSI, voltage deviation (voltage profile) and PLCC, but hydrogen storage and FCEVs create more favorable conditions for the economic, environmental and energy losses states of the network. Therefore, it is expected that the combination of resources, vehicles and hydrogen storage in IES (Case V) has better conditions for the aforementioned indicators than other cases. This is proven in Table 4. In the deterministic model, Case V has been able to reduce the operating cost, network pollution, energy losses and maximum voltage drop by about 42.5%, 62%, 43.9% and 57.6%, respectively, than in Case I. In these situations, WSI and PLCC are raised by about 26% and 51.7%, respectively. Also, Case V has been able to reduce the maximum overvoltage by roughly 47.8% than in Case II. In all cases, bus 65 has the lowest voltage magnitude. Hence, poor bus is equal to 65 in all the cases. With increasing uncertainty radius, according to Table 3, energy consumption and energy price increase, and energy generation capacity decreases. Therefore, increasing R results in increasing operating cost, energy losses and network pollution level, and voltage drop. WSI, overvoltage and PLCC decrease in this condition. In case I, it is observed that increasing R causes unacceptable conditions in the network, such that the voltage drop is more than its maximum value of 0.05 (0.95—1) p.u.

Conclusions

The optimal operation of a renewable IES with electric vehicles and hydrogen storage in a SDN was presented, considering economic, environmental, voltage security, and operation objectives of the grid operator. In the deterministic model, the objective function is equal to the minimization of the weighted sum of the operating

cost, emissions, and the grid voltage profile function. This function is constrained by the linearized optimal power flow equations, the grid voltage security limits, and the operation model of renewable resources, electric vehicles, and hydrogen storage in the form of an IES. A fuzzy decision-making algorithm was used to extract a compromise solution. This scheme has uncertainties in load, renewable power, energy price, and various parameters of electric vehicle aggregation. This paper uses adaptive robust optimization to model uncertainties in order to achieve operation that is robust against uncertainty prediction errors. Based on the numerical results, it was seen that the fuzzy decision-making algorithm was able to obtain a compromise point where the operating cost functions, network pollution and voltage profile are close to their minimum values. So, the aforementioned functions are about 23.8%, 23.3% and 21.9% away from their minimum values at the compromise point, respectively. In the worst-case scenario, the energy consumption of electric vehicles and passive loads increases than in the scenario corresponding to the deterministic model, while generation energy capacity of RESs and EVs decreases. However, energy management of IES in the distribution network has made the proposed scheme robust against a maximum prediction error of 30% for the aforementioned uncertainties. In addition, energy management of various electric vehicles and hydrogen storage devices along with RESs in the form of an IES has improved the economic, environmental, voltage security, and operation states of the distribution network by about 42.5%, 62%, 26%, and 43.9%-57.6%, respectively, compared to load flow studies.

Data availability

All data generated or analyzed during this study are included in the manuscript. Also, the datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Hadi Varei: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing—Original Draft; Hadi Zayandehroodi: Conceptualization, Methodology, Writing—review and editing, Resources, Validation, Supervision; Mehdi Jafari Shahbazzadeh: Writing—review and editing, Validation, Data Curation, Supervision; Mahdiyeh Eslami: Writing—review and editing, Validation, Data Curation, Supervision.

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Declarations

Competing interests
The authors declare no competing interests.

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